

Automatic Alcohol Sprayer



Introduction

The Automatic Alcohol Sprayer is designed to address the need for improved hygiene and reduced cross-contamination in public and shared spaces such as classrooms, offices, and entrances. Traditional manual alcohol dispensers require physical contact, which can contribute to the spread of bacteria and viruses, especially in high-traffic environments.

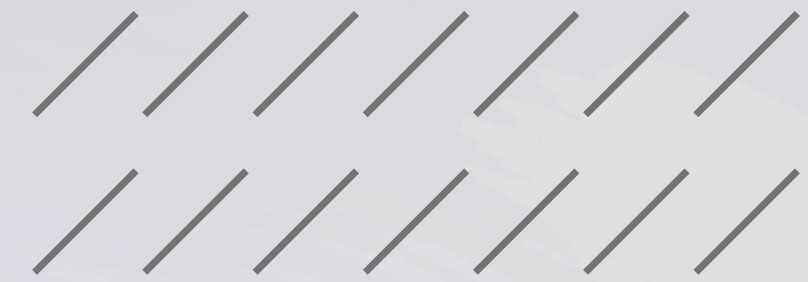


Introduction

While commercial automatic dispensers already exist, they are often expensive. This project uses an Arduino-based embedded system with an ultrasonic sensor to detect the presence of a hand and automatically activate a small pump to dispense alcohol, demonstrating how embedded systems integrate hardware and software to perform real-time sensing and control tasks.



Objectives



The goal of this project is to design and implement a touchless alcohol dispensing system that automatically releases sanitizer when a hand is detected, promoting hygiene and reducing physical contact in shared environments.

Specific objectives:

- Detect the presence of a hand using an ultrasonic sensor.
- Automatically activate a mini water pump to dispense alcohol without physical contact
- Control the system using an Arduino microcontroller as the embedded processing unit.
- Reduce the risk of cross-contamination compared to manual dispensers.
- Develop a cost-effective and energy-efficient prototype suitable for school or public use.

Scope

The project involves designing and developing a prototype Automatic Alcohol Sprayer using an Arduino microcontroller, ultrasonic sensor for hand detection, and mini pump for automatic alcohol dispensing. It focuses on basic touchless detection and automated dispensing in a small-scale setup for demonstration purposes, incorporating hardware integration and simple embedded programming. Advanced features like IoT connectivity, mobile app integration, battery optimization, liquid level monitoring, and commercial-grade casing are excluded.

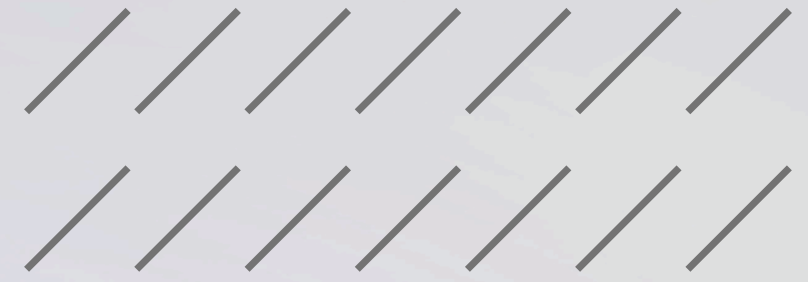
System Design

Diagram: Ultrasonic Sensor → Arduino Microcontroller → Relay/Transistor → Water Pump → Alcohol

Hardware: Arduino Uno microcontroller, HC-SR04 ultrasonic sensor for hand detection, mini 5V submersible water pump for alcohol dispensing, relay module (or transistor) for pump control, external power supply, jumper wires, and alcohol container.

Software: Arduino IDE for development, programmed using C/C++ language, utilizing digital input/output control and distance calculation logic based on ultrasonic pulse timing.

Technical Requirements



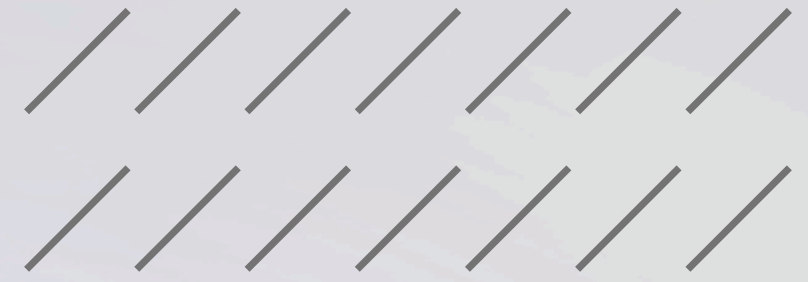
Hardware

[Arduino Uno or Arduino Nano, HC-SR04 Ultrasonic Sensor, 5V Mini Submersible Water Pump, Relay Module or NPN Transistor (e.g., TIP120), External 5V–9V Power Supply, Jumper Wires, Breadboard, Alcohol Container, Tubing]

Software

[Arduino IDE, C/C++ Programming Language, Built-in Arduino Libraries for Ultrasonic Sensor Control and Digital I/O Functions]

Technical Requirements



Tools

[USB Cable for Programming,
Multimeter for Circuit Testing,
Soldering Iron (optional for permanent
connections), Proteus or Tinkercad for
Simulation (optional), Screwdrivers
and Basic Hand Tools for Assembly]

**Thank
You!**

